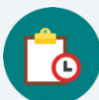
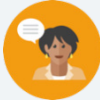


13.11 Calculating and Interpreting Average Rate of Change

Lesson Introduction

 Benchmark	MA.912.F.1.3 Calculate and interpret the average rate of change of a real-world situation represented graphically, algebraically, or in a table over a specified interval.		 Learning Targets	- I can calculate the average rate of change of a real-world situation represented graphically, algebraically, or in a table over a specified interval. - I can interpret the average rate of change in a real-world context.
 Benchmark Coherence	Building On		Working Toward	
	N/A		N/A	
 Essential Question	How do I calculate the average rate of change for a real-world situation?			
 Lesson Narrative	In this lesson, students will learn how to calculate the average rate of change for a real-world situation that is represented graphically, algebraically, or in a table. Students will be introduced to new vocabulary such as a secant line and the average rate of change, which will help reinforce their understanding of rate of change.			
 Component Summary	13.11.1 Warm-Up (~5 min.)	Critical thinking creating squares activity (<i>SE p. 291</i>)	13.11.3 Try It (10-15 min.)	Independent practice with average rate of change (<i>SE p. 296</i>)
	13.11.2 Guided Instruction (15-20 min.)	Introduction to average rate of change (<i>SE p. 292</i>)	13.11.4 Wrap-Up (~5 min.)	Ticket Out the Door on the average rate of change (<i>SE p. 297</i>)
 Resources	Glossary		Materials	
	<u>Key Terms:</u> - Secant line - Average rate of change <u>Additional Terms:</u> N/A		(Optional) Toothpicks for 13.11.1 Warm-Up	
			Framer Model, living glossaries, reference guides, anchor charts, word walls used for glossary terms.	

 Teacher Tips	Common Misconceptions - Students may make calculation errors when calculating the average rate of change. - Students may think the average rate of change is how much the function will change at every interval.	Things to Consider Students are introduced to the vocabulary term “secant line” in this lesson. It’s an introduction and shouldn’t be something that a lot of time is spent on, as they will explore more of this in both Geometry and Algebra 2.
	Literacy and Language Routines Compare and Connect 13.11.2 Guided Instruction provides students with instruction on the average rate of change. There are many instances where students may misunderstand this concept. Leverage this routine to share your thinking out loud. This supports metacognitive and metalinguistic awareness.	Mathematical Thinking and Reasoning (MTR) Standard MA.912.MTR.7.1: Real-World Contexts This lesson consists solely of real-world problems. During this lesson, students will connect mathematical concepts to everyday experiences. These problems will provide opportunities to create models, both concrete and abstract, and perform investigations.
	 Differentiate Instruction Students who are struggling readers may find this lesson to be a challenge due to the amount of text and reading comprehension necessary. Consider differentiating the problems using reading and word-problem strategies that are most appropriate for your students and their needs.	
Lesson Notes:		

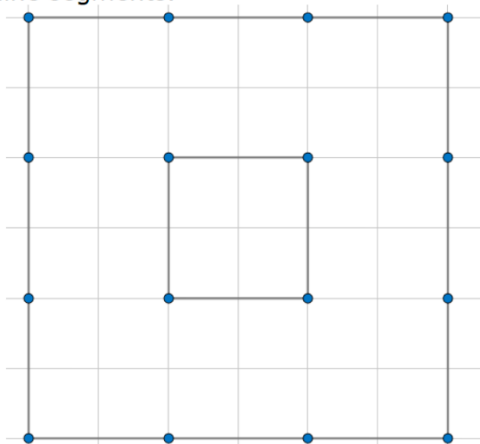
13.11.1 Warm-Up: Critical Thinking

Component Overview: Students will engage in a critical-thinking activity where they are asked to move four line segments to create three squares.

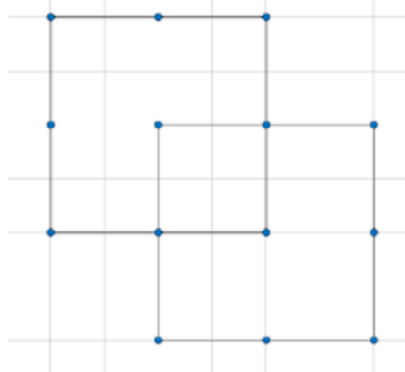


13.11.1 Warm-Up: Critical Thinking

- Two squares are shown on the grid that were created using line segments.



Move 4 line segments to create 3 squares.



Consider providing students with physical materials to provide them with an opportunity to manipulate the squares to try multiple approaches. For example, the use of toothpicks can allow students to count their moves and explore their thinking in real time to find the correct answer.

13.11.2 Guided Instruction: Average Rate of Change

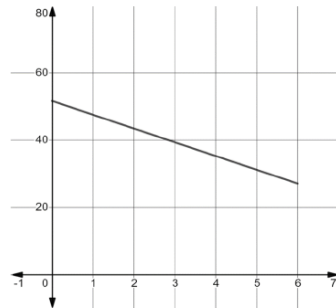
Component Overview: Students will be introduced to the average rate of change through five multipart questions.



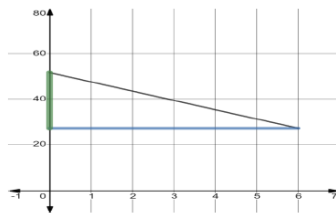
13.11.2 Guided Instruction: Average Rate of Change

1. In 2005, the percentage of the population of the Democratic Republic of the Congo living at or below the International Poverty Line was 51.6. The table and the graph show the percentage of the population from 2005 to 2011, where x is the number of years since 2005 and y is the percentage of the population living at or below poverty level.

Congo 2005 – 2011	
x	y
0	51.6
1	47.5
2	43.4
3	39.3
4	35.2
5	31.1
6	27



- a. Using the graph of the relationship, discuss with your partner how the triangle formed shows the slope.



The triangle is showing the rise and the run. It shows the decrease in y and the increase in x .

- b. Complete the table to find the slope.

Green Line Segment	$\Delta y = 27 - 51.6 = -24.6$	$\frac{\Delta y}{\Delta x} = \frac{-24.6}{6} = -4.1$
Blue Line Segment	$\Delta x = 6 - 0 = 6$	

- c. Interpret the slope in terms of the context.

Every year from 2005 to 2011, there was a 4.1% decrease per year in the population of Congo.



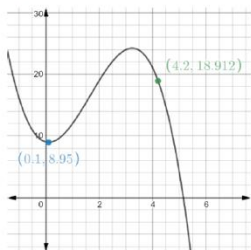
For this activity, consider beginning by providing the narrative language of question 1 for students. Use this time to prompt students to share things they notice about the table of values and the graph. Then release students to complete questions 1a-c with their partner.



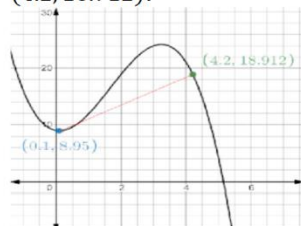
Students may not recall the symbol for change (delta). Consider asking students what it means and, if necessary, add it to your word wall or other vocabulary tool.

13.11.2 Guided Instruction: Average Rate of Change (continued)

2. The graph of a function is shown. Points $(0.1, 8.95)$ and $(4.2, 18.912)$ are on the function.



- a. Draw a line that connects the points $(0.1, 8.95)$ and $(4.2, 18.912)$.



The line drawn is a secant line.

A **secant line** is a line that passes through two points of a curve.

- b. Find the slope of the secant line that passes through the points $(0.1, 8.95)$ and $(4.2, 18.912)$.

$$\frac{9.962}{4.1} \approx 2.43$$

The **average rate of change** is the slope of the secant line. It is the change in the y -values divided by the change in the x -values for two distinct points.



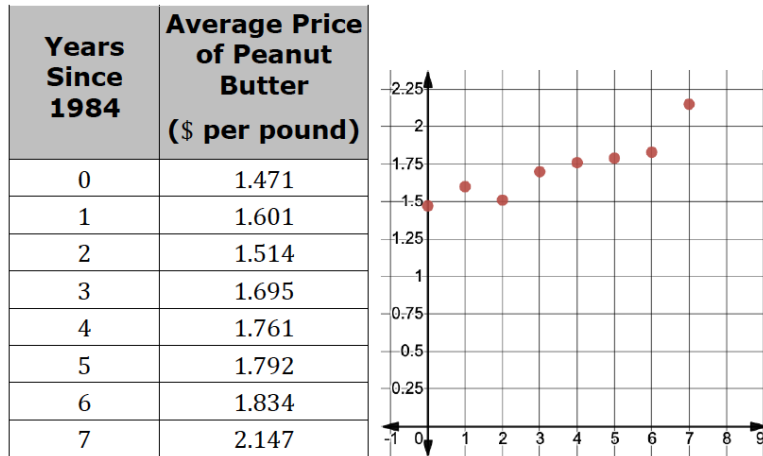
Students are introduced to the vocabulary term “secant line” in this lesson. It’s an introduction and shouldn’t be something that a lot of time is spent on, as they will explore more of this in both Geometry and Algebra 2. Add this language to any existing vocabulary models that already exist (Fayer Model, living glossaries, reference guides, anchor charts, word walls, etc.).



Do you think that the average rate of change will result in the amount of change that exists on every interval? Why or why not?

13.11.2 Guided Instruction: Average Rate of Change (continued)

3. The average price for 16 ounces of creamy peanut butter for the years 1984 to 1991 is shown in the table. The values are plotted on the coordinate grid as shown.



- a. Find the average rate of change for the interval 1985 to 1989. Include units.
 $\frac{0.191}{4} \approx 0.05$
\$0.05 increase per pound of peanut butter each year from 1985 to 1989.
- b. Find the average rate of change for the interval 1984 to 1991.
 $\frac{0.676}{7} \approx 0.10$
- c. Interpret the average rate of change for the interval 1984 to 1991 in terms of the context.
The average price of peanut butter increased, on average, \$0.10 per pound each year from 1984 to 1991.
- d. What do you notice about the average rates of change for 1985 to 1989 and 1984 to 1991?
The average rate of change from 1984 to 1991 is two times the rate of change from 1985 to 1989.



When working on question 3, help students make the connection between what they know about rate of change from prior units and the average rate of change that they are learning here.

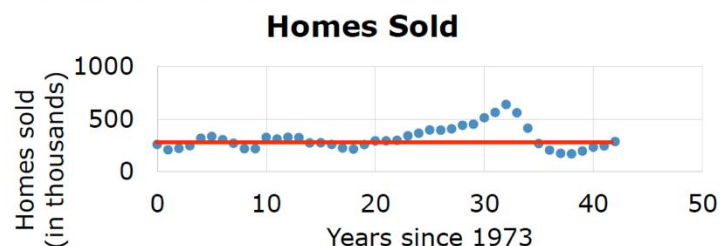


Spend some time on part c of this question. Many students will misinterpret the average rate of change using language that better represents a constant rate of change, which is not the same. The language students use here will be very important.

13.11.2 Guided Instruction: Average Rate of Change (continued)

4. Mathematicians and economists often want to evaluate changes that have occurred over different time intervals to analyze trends and make predictions.

The graph shows the number of homes sold in the U.S., in thousands, from 1973 to 2015.



- Draw a line from the first data point to the last data point.
- Discuss with your partner why finding a rate of change using the first data point to the last data point would not give much information about how the number of sales has changed.

You are using two points that are both almost level and not considering the changes in the curves of the graph.

The table shows some of the values from the graph.

Years Since 1973	Homes Sold (in thousands)	Years Since 1973	Homes Sold (in thousands)
20	292	32	639
21	292	33	559
22	297	34	412
23	340	35	266
24	364	36	203
25	396	37	172
26	394	38	168
27	406	39	194
28	440	40	232
29	451	41	244
30	511	42	285
31	562		

?

What are some pieces of information that we can gather from this data display?

?

How can the spread of values used when finding the average rate of change play a role on the outcome of the calculations and interpretation?

?

Discuss with your partner why that time period may be of interest based on the graph. Turn and talk.
What is another time period that may be of interest?

13.11.2 Guided Instruction: Average Rate of Change (continued)

- c. Find the average rate of change for the time period of 1993 to 2005.

$$\frac{347}{12} \approx 29$$

- d. Interpret the average rate of change for the interval 1993 to 2005 in terms of the context.

There is an increase of 29 thousand homes sold each year from 1993 to 2005.

5. The function $M(x) = 5x^2 - 160x + 6179$ models the number of miles, in millions, U.S. passengers traveled on a train for the period of 1970 to 2000, where x is the number of years since 1970.

- a. What is the average rate of change, in miles traveled by U.S. passengers per year, for the period from 1986 to 2000?

70

- b. What is the average rate of change, in miles traveled by U.S. passengers per year, for the period from 1970 to 1980?

-110

- c. Discuss with your partner what the slope indicates about whether the graph is increasing or decreasing on the given interval.

If it's a positive value, there is an increase. If it's a negative value, there is a decrease.



Consider releasing students to work with a partner on question 5a. Then have them rotate to a new partner for question 5b. Then have them switch to a new partner again for question 5c, having to calibrate their answers with each new partner. Finally, have them complete question 5c again with a different partner for a total of four student pairings.

13.11.3 Try It: Average Rate of Change

Component Overview: Students will complete two practice questions independently on finding the average rate of change.



13.11.3 Try It: Average Rate of Change

- Mrs. Wilson's class conducts an investigation to find out how quickly the height of water in a half-gallon jug changes after a small hole is created near the bottom of the jug. The class finds the function $H(x) = 0.01x^2 - 2.77x + 136.51$ models the height, in millimeters, of water x seconds after the hole is created.

What is the average rate of change, in millimeters per second, for the period of 37 to 53 seconds? Round your answer to the nearest hundredth.

-1.87 mm/s

- The table shows the number of tons of blueberries that were imported to the US from 2005 to 2013.

Year	Blueberries Imported (tons)
2005	26,319
2006	32,602
2007	35,125
2008	52,111
2009	60,450
2010	76,743
2011	87,336
2012	96,367
2013	103,502



Consider using the time in this activity to work with a small group of students who performed poorly during the 13.11.2 Guided Instruction activity.

- Determine the average rate of change from 2007 to 2012.
12,248.4
- Interpret the average rate of change for the interval 2007 to 2012 in terms of the context.
There was an increase of 12,248.4 tons of Blueberries imported each year between 2007 and 2012.

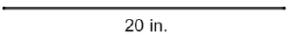
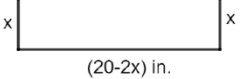
13.11.4 Wrap-Up: Ticket Out the Door

Component Overview: Students will complete a Ticket Out the Door that assesses their understanding of average rate of change from real-world scenarios. Students should be allowed to access a calculator to complete the Wrap-Up.

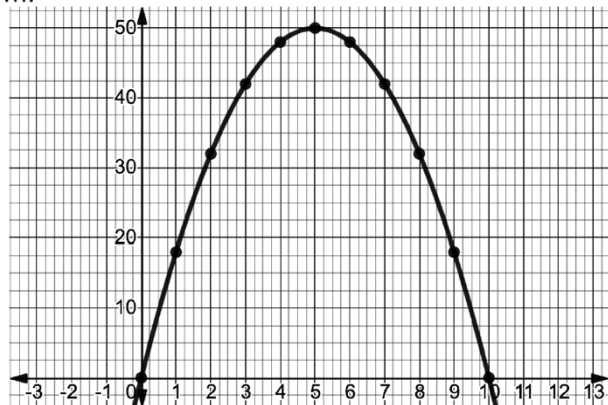


13.11.4 Wrap-Up: Ticket Out the Door

A rain gutter is made from a 20 inch (in.) aluminum sheet with the edges folded up, as shown.

Flat Aluminum Sheet	Sheet Folded for Rain Gutter
 20 in.	 (20-2x) in.

The function $A(x) = x(20 - 2x)$ models the cross-sectional area of a rain gutter made from the sheet, where x is the height of the rain gutter. The graph of the function is shown.



- What is the average rate of change for the interval of heights of 1 in. to 5 in.
8 square inches per inch
- Interpret the average rate of change for the interval of 1 in. to 5 in. in terms of the context.
There is an 8 square inch increase in the areas for each inch added to the height of the sheet.



Consider our online resources for additional enrichment opportunities for students to practice or test their abilities.